The prosody of non-canonical questions in Swedish

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1 Introduction

This chapter gives an overview over the body of work on the prosody of non-canonical questions in Swedish, with a special focus on experimental and corpus studies. Swedish prosody in general is quite well-studied: The presence of tone accents in most varieties of Swedish captured the attention of researchers from the earliest days, and Gösta Bruce's seminal account of tone accents in Bruce (1977) proved to be an important building block in the development of autosegmental-metrical phonology (cf. the historical overview in Ladd, 2008), and, importantly for us, highlighted the need to investigate sentence prosody (i.e. prosodic distinctions between e.g. statements and questions) even for research focusing on word prosody (i.e. the tone accents).

Swedish makes for an interesting case study with respect to the prosody of questions, particularly that of non-canonical ones, because it appears to be settled at this point that the tone accents are stable in their phonological identity across different sentence types, including statements and questions (e.g. Riad, 2018, p. 383). However, there seem to be fairly consistent *phonetic* modifications of tone accents in questions, such as expanded pitch range and delayed pitch peaks (Gårding, 1979; House, 2002, 2004), and non-pitch related differences such as duration (House, 2003). An interesting question to ask is whether those phonetic modifications are stronger in non-canonical questions – e.g. because nothing besides these modifications marks them as questions – or perhaps weaker – because non-canonical questions are less 'question-like' than canonical questions.

First, some typological background on Swedish: Swedish is a North Germanic verb-second SVO language in which canonical polar questions are formed by inversion of the finite verb and the subject, resulting in verb-first polar questions, (1b), while *wh*-questions are formed by fronting the *wh*-phrase to the pre-verbal position, (1c). If a *wh*-phrase is left in-situ, the question functions as an echo question, (1d). In other words, the general properties of canonical question formation work much like in closely related Germanic languages. In addition to these question types, Swedish has questions with declarative syntax (henceforth *declarative questions*), which can optionally contain modal particles and/or be followed by question tags. (1a) is a declarative sentence that is in principle ambiguous between a statement and a question, indicated here by punctuation.

(1) a. Per har träffat LINA./?
P. has met L.
"Per has met Lina./?"

b. Har Per träffat LINA?
has P. met L.
"Has Per met Lina?"

c. Vem har Per TRÄFFAT?
who has P. met
"Who has Per met?"

d. Per har träffat VEM?
P. has met who
"Per has met who?"

Nuclear accent placement and focus projection are very similar to related Germanic languages such as English. I follow the definitions outlined in Myrberg (2021a): The nuclear accent is defined as the final strong accent in an intonation phrase (see Section 2.1 for an elaboration of "strong"). A narrowly focused word attracts the nuclear accent. In broad(er) focus, the nuclear accent is generally realized as far to the right as possible within the intonation phrase corresponding to focus. I have indicated the nuclear accents in (1) with small caps. (1a) could be used as an answer both to "What happened?" (in which case there is focus projection, because the accent on *Lina* marks focus on the entire sentence), as an answer to "Who did Per meet?" (in which case there is narrow focus on *Lina*), or as a correction to "Per met Mona" (in which case there is narrow and contrastive focus on *Lina*).

Turning to the definitions of "prosody" and "non-canonical questions", prosody is here understood to include tone (grammaticalized, lexical uses of pitch[1]), intonation (post-lexical uses of pitch), stress, rhythm, intensity, and voice quality. However, the vast majority of research on Swedish question prosody is exclusively focused on pitch. Regarding the canonical vs. non-canonical question distinction, I take a narrow view of 'canonical': Canonical questions are here understood to only include syntactically marked (i.e. *wh*-initial *wh*-questions and verb-first polar questions), fully sentential, and neutrally information-seeking questions. In other words, a question is non-canonical if it has declarative syntax, a non-fronted *wh*-word, or is fragmentary; and it is non-canonical (even if it is syntactically marked and sentential) if any formal aspect of it conveys another function in addition to information-seeking (such as surprise, incredulity, mirativity, previous beliefs of the speakers, bias, etc.) or a primary function *other than* information-seeking (in the case of rhetorical questions, deliberative questions and reported questions).

This chapter is structured as follows. Background information on Swedish tone and intonation in general is presented in Section 2. Section 2.1 focuses on statement intonation, in order to introduce the tone accents in the context in which they have been most thoroughly studied. Section 2.2 gives a top-level overview of what we know about the prosody of Swedish questions. Section 3 highlights a few experimental and/or corpus studies, particularly of non-canonical questions. Section 4 discusses a potential 'conflict' between sentence prosody and word prosody. Section 5 concludes.

[1]Pitch is the psychoacoustic correlate of the fundamental frequency of the voice, *F0*.

2 Word and sentence intonation of Swedish

2.1 Statement intonation

One of the most salient prosodic features of most varieties of Swedish is the presence of tone accents.[2] These varieties distinguish between two accents, generally called *accent 1* and *accent 2* (sometimes *acute accent* and *grave accent*; henceforth A1 and A2). *Equipollent* analyses of Swedish tone accents analyze both A1 and A2 as lexically specified properties of individual words and morphemes. Bruce (1977) provides such an equipollent analysis. Other analyses are *privative*, in which only one of the two tone accents is lexically specified, while the other is treated as the absence of lexical tone and its surface form is the result of post-lexical processes. The lexically specified tone is usually assumed to be A2 (but see Lahiri, Wetterlin, and Jönsson-Steiner, 2005, for a privative analysis in which A1 is lexically specified). An important feature of the Swedish (and Norwegian, cf. Riad, 2018; Myrberg, 2022) tone accent system is that the concrete realizations of A1 and A2 vary substantially between dialects.

To get an impression of this variety, consider Fig. 1, which shows schematized F0 contours for the citation forms of disyllabic words with initial stress (after Gårding and Lindblad, 1973). Based on the shape of A2, the dialects can be grouped into single-peaked varieties (Southern Swedish and Dalarna/Gotland) and double-peaked varieties (Central and Western Swedish). Within these groups, Southern Swedish and Central Swedish have relatively earlier alignments of pitch peaks for both accents than the other member of each group. Finally, Finland Swedish does not have a tone accent distinction – both A1 and A2 words are realized with a single pitch peak on the stressed syllable, with no differences in peak alignment. See Gårding and Lindblad (1973), Bruce and Gårding (1978), Bruce (2007, 2010), and Riad (2006, 2018) for more on tone accent typology in Swedish (and Norwegian).[3]

This brings us to Bruce (1977), who investigated the word and sentence prosody of Stockholm Swedish, a Central Swedish variety with double-peaked A2 (row 4 in Fig. 1). Bruce proposed that the second peak in A2 words and the only peak in A1 words are phonologically the same object: a high (H) sentence accent (later called focus accent). In fact, he proposed that the entire tonal specification of A1 and A2 words in citation form was identical: HLH. The differences in surface shape were argued to be the result of different *associations* of the tonal specification with the stressed syllable. In A1, the association is HL* H, i.e. the lexical L tone is aligned with the onset of the stressed vowel and the contour rises throughout the stressed vowel to the sentence/focus accent, with the post-tonic fall being explained by a low boundary tone. In A2, the alignment is H*L H, i.e. the lexical H tone is aligned with the onset of the stressed vowel, which leads to a falling contour in the stressed vowel before the post-tonic rise-fall induced by the high sentence/focus accent and low boundary tone.

As has already been noted, the ideas of tonal targets having different origins

[2]Alternative terms include *word accents*, *pitch accents* and *lexical accents*.

[3]Note that the grouping into single- and double-peaked and early- and late-aligned varieties is the traditional grouping. See Riad (2018) and Myrberg (2022) for recent proposals that do away with peakedness and timing as theoretical primitives.

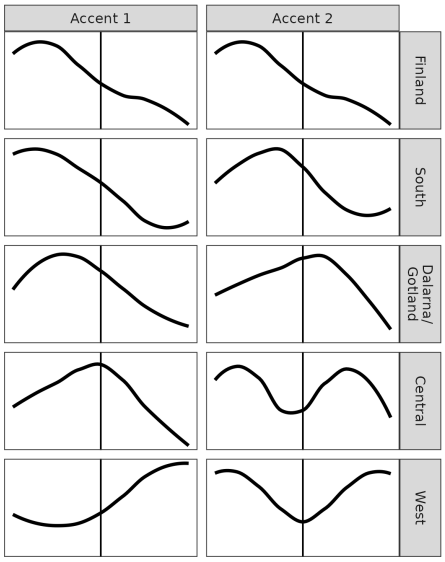

Figure 1: Accent variation across Swedish varieties, after Gårding and Lindblad (1973, p. 48)

(lexical and post-lexical, ≈ tone and intonation) and of differing associations of those tonal targets to the segmental level proved to be an important building block in the development of autosegmental-metrical phonology (starting with Pierrehumbert, 1980). However, today the majority view regarding North Germanic tone accents, including Stockholm Swedish, is that a privative analysis, in which only A2 contains lexical tone, is to be preferred (e.g. Riad, 2006; Myrberg and Riad, 2015; Riad, 2018).

Two more terms need to be introduced before our picture of (Stockholm) Swedish sentence prosody is complete: big accent and small accent. These terms were introduced by Myrberg and Riad (2015) to serve as more theory-neutral replacements for the earlier terms "sentence accent" or "focus accent" (replaced by big accent) and "word accent" (replaced by small accent). Fundamentally, the distinction between big accents and small accents is one of prominence levels. Big accents are more prominent (and contain more tonal targets in the case of A2) than small accents. Making statements about the *functions* of big and small accents is more complicated, because there is no one-to-one mapping between their form and function. For our purposes, it suffices to note that nuclear accents are big accents (with very few exceptions, cf. Ambrazaitis, 2009; Myrberg, 2010), that postnuclear accents are small accents, and that prenuclear accents also tend to be small, with the notable exception of what Myrberg (2010) termed *initiality accents*, which are prenuclear big accents that are aligned with the left edge of phrases and serve as phrasing cues (see Myrberg, 2010, 2021a).

With this vocabulary now in hand, the tone accent inventory of Stockholm Swedish is given in Table 1. Within small accents, the difference between A1 and A2 is one of association of an HL tone pair with the stressed syllable, indicated by the location of the star. Within big accents, the difference is whether

a post-lexical LH tone pair – the *prominence tone* (Riad, 2006) – can associate with the stressed syllable (A1), or is 'pushed' out of the stressed syllable by the lexical H tone, which associates with the syllable instead (A2). Finally, in Stockholm Swedish compounds, the H* tone is also post-lexical (because all Stockholm Swedish compounds have A2, even if all component words have A1 and thus no lexical H* tones), and the post-lexical L tone associates with the final stressed syllable in the compound, again indicated by the star.

Table 1: Tone accents of Stockholm Swedish according to Riad (2018), building on Myrberg and Riad (2015). Lexical tones are in bold face.

	A1	A2 (simplex)	A2 (compounds)
Small accent	HL*	**H***L	H*L
Big accent	L*H	**H***LH	H*L*H

The final concept we need for a complete account of Stockholm Swedish statement intonation (and sentence intonation, more generally) are boundary tones. For most Stockholm Swedish statements, the boundary tone is low and notated L%. I will discuss boundary tones in more detail in connection with question intonation in Section 3.1.1.

2.2 Question intonation

Turning now to the intonation of Swedish questions, consider the following quotation from Gårding (1979), which summarizes two things very well: the traditionally-assumed connection between intonation and sentence type cross-linguistically, and the types of theoretical questions that Swedish specifically raised for researchers of question intonation:

> Sentence intonation for practically all languages is usually described as 'falling' for statements and 'rising' for yes/no questions.
>
> I take sentence intonation to mean tonal features signalling sentence type (e.g. statements and questions of various kinds) with correlates in the fundamental frequency curve. With this definition in mind, it is natural to ask: What is it that falls and rises in an accent language like Swedish, where pitch (fundamental frequency) is constantly forced to go up and down by the different accents? Is the rise the impression left by an overall higher fundamental frequency? Or is it the result of a local affair at the end, at the beginning, or at some other place in the utterance?
>
> (Gårding, 1979, p. 207)

The aim of Gårding (1979) was to develop a combined model of word and sentence intonation for four Swedish varieties (South, Dalarna/Gotland, Central and West, i.e. the main four varieties of the tone accent typology), so these theoretical questions needed concrete answers. The answers given by Gårding are a compromise between the two options that she presented: F0 is higher in Swedish questions than in statements on average, but this global difference is the result of local differences. Specifically, she suggests that the F0 range related to the sentence accent is wider in questions than in statements. In current

terminology, this corresponds to a raising of the H tone of the prominence tone belonging to the nuclear accent.[4]. Put more descriptively, a high target in the post-tonic syllable is raised in questions for three out of the four dialects that she models (the exception is South Swedish, where the raised high target is inside the tonic syllable). Note that Gårding (1979) only discusses question intonation for (fragmentary) phrases consisting exclusively of A2 words. For A1 nuclear accents, which are aligned earlier across varieties than A2, we should probably expect more differences on the tonic syllable itself (but either way, a modification only of the post-lexical prominence tone).

Gårding (1979) is particularly relevant for the present chapter because Gårding models statements, canonical verb-first yes/no questions and non-canonical verb-second declarative questions separately for South Swedish (Gårding, 1979, p. 213, Fig. 4).[5] Fig. 2 is a recreation of the final output of Gårding's model. The differences between the statement and both questions are: (i) the pitch peak on the stressed syllable of the nuclear accented word is raised, (ii) the prenuclear accents have higher lows, i.e. the prenuclear 'baseline' is raised, and (iii) there is a rise at the end of the depicted contours, which rises to a height that is in line with the prenuclear accents, allowing for declination (note that also the statement has a small rise). The difference between the verb-first question and the verb-second declarative question is that the latter has an even higher nuclear accent pitch peak and that the prenuclear accents are compressed in an overall higher pitch register, by further rising of the lows between accents. Overall, there is also less declination in the declarative question, i.e. there is less downstep between prenuclear accents.

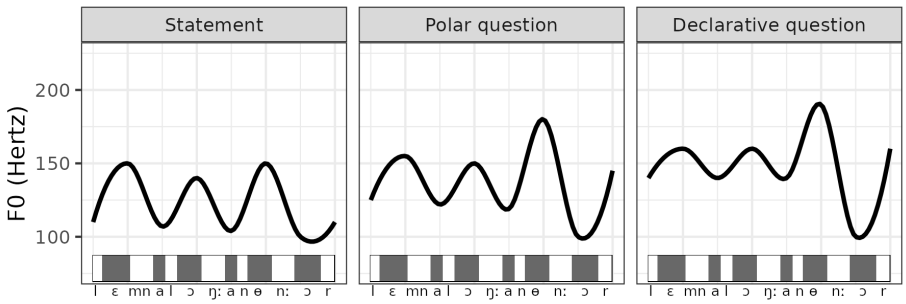

Figure 2: Modeled intonation contours for one Southern Swedish speaker, after Fig. 4 in Gårding (1979, p. 213). The spoken phrase is *lämna långa nunnor*, "leave long nuns". All words have A2 and the nuclear accent is on *nunnor*.

An interim summary: Swedish question intonation as modeled by Gårding (1979) consists of phonetic *modifications* of statement intonation: certain high pitch targets that exist also in statements are relatively higher in questions, and there is a raising and compression of prenuclear pitch range. Gårding summarizes this by saying that "the dialectal variation in question intonation hinges on the different dialectal manifestation [sic!] of sentence accent [≈ prominence tone]" (Gårding, 1979, p. 214). Based on Fig. 2, it would appear

[4]Or of the boundary tone H% for dialects which have a L prominence tone, cf. Riad, 2018, p. 354

[5]The speaker for these was Gösta Bruce, who was from Helsingborg in Southern Sweden.

that there is a continuum of question intonation, in that the non-canonical declarative question receives slightly more of the same modifications that the canonical polar question receives. Interestingly, question intonation seems to most strongly phonetically modify *post*-lexical tones, even in those dialects that have a H lexical tone in A2 (Central and Western Swedish, cf. Riad, 2018).

To close this section, I want to give some examples of non-modeled F0 contours. Fig. 3 shows four F0 contours of utterances spoken by one Central Swedish participant of the production experiment reported in Seeliger and Repp (2017). The experiment manipulated two factors: the intended speech act (rejection or rejecting question) and the location of contrastive focus (verb or object). A rejecting question can be used to indicate a desire to reject something, without having the epistemic authority to outright reject it. (2) gives translations of the context sentences and the target sentence for rejecting questions with contrastive focus on the verb. The object focus conditions introduced a focus alternative for the object ("sheep"), and the rejection conditions contained continuations in which the speaker was certain about what they were rejecting. The context sentence C and the contribution of S1 were pre-recorded, while the contribution of S2 was only shown in written form and spoken by the participants. Note that the four words of the underlined target utterances had A2 for all items.

(2) C: A conversation about Mona, a well-known cattle merchant.

S1: Mona has recently started selling llamas. But I get the impression that she has stolen them.

S2: <u>Inte</u> <u>nallar</u> <u>Mona</u> <u>lamor</u>? Hon är ju så framgångsrik. Hon borde
not steals M. llamas she is MP so successful she should
väl ha råd att köpa dem?
MP be.able.to.afford to buy them
"Surely Mona doesn't steal llamas? She is so successful. Surely she should be able to afford to buy them?"

Differences between the speech acts are clearest for the verb focus condition in Fig. 3: in the rejecting question, there is a substantial raising of pitch on the focused verb, especially on the second peak of the verb (i.e. the H of the prominence tone). In the object focus condition, differences are smaller, but there is again raised pitch both on the prenuclear initiality accent on the verb (which merges with the H*L small accent on the subject, cf. Bruce, 2010, pp. 88–89, Myrberg, 2010, p. 57) and on (the prominence tone of) the nuclear accent on the object. This speaker has a strong tendency for utterance-final creaky voice, which somewhat masks the audible final fall to L% in all conditions. Overall, it can be seen that the intonation in these particular non-canonical questions accords quite well with the description in Gårding (1979).

In a nutshell, the answer to the question *What is the prosody of questions in Swedish?* has remained largely unmodified since Gårding (1979). There are, however, some open issues. The issue that has occupied a central position in the literature is to what extent Swedish questions are characterized by utterance-final rises, or to what extent the rises that do occur are question-forming. Another issue, more central to this chapter but rarely brought to the forefront in the literature, is to what extent non-canonical questions share

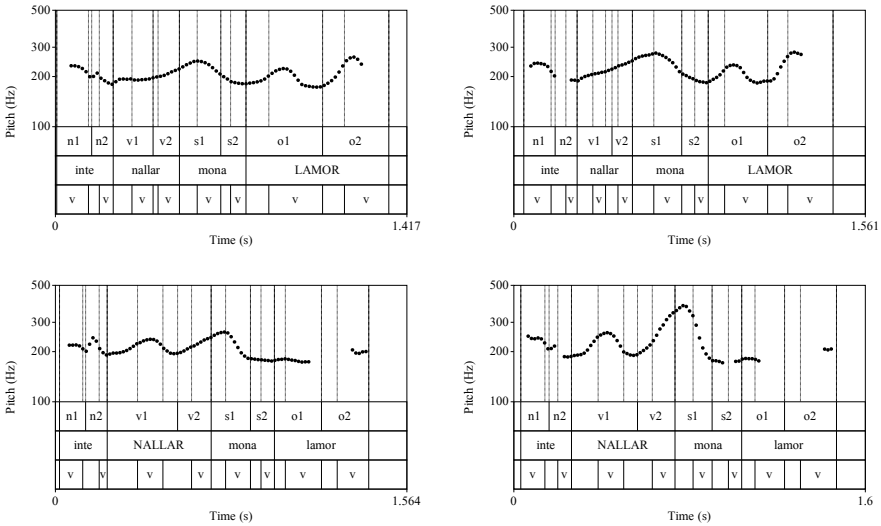

Figure 3: Four productions of the sentence *Inte nallar Mona lamor* ('Mona doesn't steal llamas') by one Central Swedish speaker (Seeliger and Repp, 2017). Left: rejections. Right: rejecting questions. Top: object focus (final). Bottom: verb focus (medial). Tiers are from top to bottom: syllables, words (contrastive focus in capitals), vocalic intervals. /r/ in codas has been treated as vocalic, and syllable boundaries involving geminates have been annotated in the middle of the consonant.

prosodic properties with canonical questions. Do non-canonical questions exhibit more "extreme" question prosody than canonical questions, because the latter are already syntactically marked?

2.3 What's in a rise?

Before we proceed to the discussion of experimental studies, the term "(intonational) rise" needs to be defined. Rising intonation (or just high pitch) is often connected to question intonation cross-linguistically (cf. Gussenhoven, 2004, pp. 80–84). This is also the case in discussions of Swedish prosody. Unfortunately, the meaning of "rise" is not always kept completely clear. At least three different meanings of "rise" can be identified in the literature on Swedish question prosody:

1. **Question rises**: This type of rise is easiest to define by making reference to English. Where many unmarked English statements might have an entirely post-lexical intonation contour transcribable by H* L-L% (high nuclear accent followed by a fall to utterance-final low pitch), many unmarked yes/no questions might have contours transcribable by L* H-H% (*low* nuclear accent followed by a *rise* to utterance-final *high* pitch). We have not seen any evidence so far that Swedish has such a stark distinction between "statement contours" and "question contours" (and in fact there are explicit claims that it does not, e.g. Riad, 2018, p. 383), but

this type of rise keeps surfacing in the literature on Swedish. For example, Gårding and Lindblad (1973, pp. 83–84) explicitly draw a connection to other Germanic languages: "The [South Swedish] speaker alone marks her questions by reversing the intonation patterns of the statements. In this way, the South Swedish speaker behaves similarly to speakers of other Germanic languages such as Danish, German and English".[6]

2. **Tone accent rises**: In some configurations, tone accents can result in very salient utterance-final rises. For example, West Swedish and East Norwegian generally have rising intonation in statements, which Riad (2018) analyzes as the result of a phrase-final H% boundary tone. This boundary tone can spread backwards all the way to the L prominence tone and 'stand in' where other varieties have a H prominence tone (followed by a L% boundary tone in those varieties). Another example are some of the South Swedish double-peaked late nuclear accents analyzed by Myrberg (2022), in which the underlying contour is analyzed as L*H LH, and there is optional truncation of the final H tone. If it is not truncated, an utterance-final rise results.

3. **Non-rises**: These are cases when contours are discussed as if they were rising in one of the first two senses, even though they are not. A clear case is in Gårding (1998, p. 117), which shows F0 contours of (South Swedish) sentences containing tone accent pseudo-minimal pairs in prenuclear position, which were uttered as a statement, as a question and as an order. There are certainly some differences between the different sentence types, but they all end in a fall to low pitch after the A2 nuclear accent. Nevertheless, Gårding writes that the questions are "characterised by a continuous rise", which "is most clearly expressed by the topline of the accent humps" (Gårding, 1998, p. 122). While the reference to the topline does make it clear that Gårding is not referring *directly* to the F0 contour (but instead to the relative heights of accent peaks), I think that a term other than "rise" would be more appropriate here (e.g. "suspension of declination" – cf. Grønnum's work on declination in Danish; Grønnum, 1998).

Based on all available evidence, I consider it unlikely that (any variety of) Swedish has phonologized "question contours", in which the actual phonological identity of the tone accents is changed because they occur in a question rather than a statement. It is clear that Swedish has tone accent rises, but these are independent of interrogativity. Finally, Swedish question prosody does indeed involve high pitch, as we have already seen, but it is arguably unhelpful to frame this relationship in terms of rises, because this suggests a much greater similarity to, e.g., English than there is evidence for.

[6]Note that the South Swedish speaker was Gårding herself.

3 Experimental investigations of Swedish question prosody

This section discusses utterance-final pitch movements (3.1), the phonetics of the nuclear accent (3.2), earlier pitch-related differences (3.3), and duration (3.4). This section is thus organized by phenomenon instead of by dialect. However, I will note for each study which dialect was investigated (to the extent that this was reported).

3.1 Utterance-final pitch movements

Historically, much attention was paid to utterance-final pitch movements when it comes to the prosodic distinction between Swedish statements and questions. Presumably, this was a result of the received wisdom that, in related Germanic languages such as English or German, it is utterance-final pitch movements that categorically distinguish between statements and questions (which is a simplification in itself; cf. Couper-Kuhlen, 2012, on English; Petrone and Niebuhr, 2014, on German). In the following sections, I will first present phonological accounts of Swedish utterance-final pitch movements (Section 3.1.1), then discuss experimental investigations of the phenomenon (Section 3.1.2).

3.1.1 Phonological accounts: Inventory of boundary tones

Phonological accounts of Swedish generally agree on the existence of two boundary tones, L% and H%. Riad (2014, p. 266) says that "[...] L% is regular in most types of utterances, including questions", while H% is said to signal continuation or "sometimes a question". Citing House (2004), he says that, when it occurs in questions, "H% is strongly correlated with polar (yes/no) questions" (note that House, 2004, exclusively focuses on *wh*-questions). Myrberg (2010, pp. 100 ff.) similarly assumes L% and H% for Stockholm Swedish, although the inventory of phrase-final falls and rises is larger than implied by this binary opposition due to phrasing distinctions. Regarding specifically interrogative intonation, she states that H% "can be used in continuation contexts, as well as in questions" (p. 104). She agrees with House (2003, 2004) in saying that a phrase-final rise is probably not an independent question marker (since it can occur in non-questions, and questions do not necessarily require it). Ambrazaitis (2009) also assumes two boundary tones for Stockholm Swedish, notated as L% and LH% (the latter of which corresponds to utterance-final rising pitch). Where he goes beyond standard assumptions is in the presence of two different sentence accents (≈ prominence tones), the 'default' high sentence accent H- and an additional low sentence accent L-. Overall, this allows him to account for the nuclear patterns early fall, medial fall, fall-rise and simple rise, all of which were produced by the Stockholm Swedish participants in his production studies.[7] He suggests that Swedish has more post-lexical intonation than was assumed at the time (e.g. Bruce, 2005, 2007).

[7] Note that the rising contours were not particularly strongly associated with questions: 6.7% of one question type (≈ auditory echo questions) had a fall-rise; all other questions exhibited falling contours. Disbelieving questions (≈ incredulous echo questions) had strikingly many early falls (Ambrazaitis, 2009, p. 94).

3.1.2 Empirical investigations of utterance-final pitch movements

House (2004, 2005) analyzed utterance-final pitch movements in 200 *wh*-questions (100 containing *vad*, "what", and 100 containing *var*, "where"). The presented examples all appear to be canonical, information-seeking questions. The questions were produced by members of the public directed towards an automatic dialogue system exhibited in central Stockholm (the data were not controlled for dialect or L1). House found that 22% of all questions ended in a final rise. Split by age and gender, the questions produced by children had the most final rises (32%), followed by women (27%) and men (17%). He also observed a correlation between final rises and final nuclear accents: 77% of the questions ending in a rise also had a nuclear accent in utterance-final position, while only 48% of the questions with a final low had a final nuclear accent. House calls a final rise an 'optional interrogative marker', going on to suggest that it might have a more general pragmatic meaning, such that a final rise is a signal of socializing, while a final fall signals information-seeking (House, 2005, p. 273). The results of two perception studies reported in House (2005) lend some support to the hypothesis that late-aligned and high pitch is associated with an impression of friendliness.

Strömbergsson, Edlund, and House (2012) built a corpus of 600 questions selected from 120 half-hour spontaneous conversations (the dialectal makeup was not reported, but the recordings took place at KTH Stockholm). Their inclusion criteria were as wide as possible: "Anything that resembles, structurally or functionally, in whole or in part, a question" (Strömbergsson, Edlund, and House, 2012, p. 840). They annotated the questions for four criteria, some of which unfortunately turned out to be problematic – e.g. rhetorical questions were categorized alongside self-directed and reported questions (in that none of them permit answers in the narrow sense), and it is unclear if declarative questions were treated as yes/no questions or "other".

The central prosodic finding of Strömbergsson, Edlund, and House (2012) is that Swedish *wh*-questions have rising intonation and yes/no questions falling intonation, i.e. the exact opposite pattern that is commonly ascribed to languages such as English or German. The way that the authors measured falling vs. rising pitch was by taking the difference between average F0 in the two halves of the utterance. The authors themselves call this a "rough estimate of intonation slope" (Strömbergsson, Edlund, and House, 2012, p. 842), but it should be noted that this measure will generate both false positives (questions that 'look' like they're rising, but actually just have a very prominent late accent followed by a fall) and false negatives (questions that indeed have a perceptible rise, but also e.g. high prenuclear pitch or a prominent early accent). It is certainly not possible to say anything about the question rise vs. tone rise distinction from this measure (in the terminology of Section 2.3). The significant difference between *wh*-questions and yes/no questions consists of the finding that yes/no questions fall in pitch by about 0.6 semitones, while *wh*-questions rise by less than 0.1 semitones (Strömbergsson, Edlund, and House, 2012, Fig. 5). These differences would probably be imperceptible. Even the *difference* of 0.7 semitones between yes/no questions and *wh*-questions is very small. A more cautious conclusion would be that both Swedish yes/no and *wh*-questions are comparatively level in pitch, i.e. I would expect statements

to fall by more than 0.6 semitones on this measure.[8]

Huhtamäki (2015a) and the component articles cited therein (Huhtamäki, 2012, 2014, 2015b,c) analyze a corpus of spontaneous conversations held in Helsinki Swedish (the variety of Finland Swedish spoken in Helsinki). Recall from Section 2.1 that Finland Swedish does not have tone accents, which means that tone accent rises cannot be relevant here. The 'tonelessness' of Finland Swedish also opens up the possibility that it has more post-lexical intonation than Sweden Swedish, which makes Finland Swedish an interesting variety to research (cf. also Kuronen and Leinonen, 2010, on the prosody of Finland Swedish *wh*-questions).

Huhtamäki's definition of 'question' is very wide and functional: a question is an utterance that requests a response from the addressee, if the addressee knows more about the topic than the speaker (Huhtamäki, 2012, pp. 157f.). Rhetorical questions and news markers are thus explicitly excluded. The focus of the articles is as follows: Huhtamäki (2012) investigates utterance-final intonation in questions in general, Huhtamäki (2014) investigates questions with non-interrogative syntax (i.e. declarative and fragmentary questions), Huhtamäki (2015b) focuses on the function of questions with rising or level utterance-final intonation, and Huhtamäki (2015c) investigates the prosody of questions consisting only of *va(d)* ('what'). Overall, rising questions were rare (17% of the corpus investigated in Huhtamäki, 2012), even in questions with non-interrogative syntax, and rising intonation appears to be mostly an interactional marker related to e.g. incredulity or the signalling of non-understanding, rather than being a question marker per se. Based on Huhtamäki's studies, it thus does not seem likely that Finland Swedish (or at least Helsinki Swedish) patterns in categorically different ways from Sweden Swedish when it comes to question intonation, despite the absence of tone accents in Finland Swedish: question intonation is rather similar to statement intonation for both canonical and non-canonical questions.

We return to Sweden for the two most recent studies. Myrberg (2021b) investigates three Southern Swedish varieties under the hypothesis that late focus and geographic proximity to Western Sweden (in which utterance-final pitch is generally rising, also in statements, cf. Fig. 1) increase the proportion of rising statements. She does indeed find utterance-final rising intonation in both statements and questions, with substantial variation between dialects and focus structures (cf. Myrberg, 2022, for a phonological analysis of the same dataset). Finally, Horne and Roll (2021) present a corpus study of one speaker from Malmö (Southern Sweden), a journalist interviewing politicians. Their corpus consists of 38 yes/no questions, 37 *wh*-questions, 37 statements and 24 utterances 'that were introduced as Y/N questions by the interviewer' (the examples that they give are fragmentary questions, e.g. *NATO membership?*). They found that 6 of the yes/no questions, 1 *wh*-question and 19 of the fragmentary questions had final rising intonation. This indicates that final rises are optional in questions produced by this particular Southern Swedish speaker, but more common in questions that do not have lexical markers of questionhood. From a functional point of view, the fragmentary questions could be viewed as reduced

[8]Johansson (1978, pp. 33, 41) reports generally falling intonation for Central Swedish fragmentary statements using the same measure, although the statements are so short that there is a confounding effect of the tone accents, as pointed out by Johansson herself (Johansson, 1978, pp. 34–35).

wh-questions (e.g. *What's your opinion on NATO membership?*). Conversely, if the answers were restricted to short *yes* or *no* answers, with the questions coming in quick succession, it is also possible that the rises were continuation rises, with the questions forming members of a list.

Overall, we find that questions with final rises certainly exist in Swedish, but a final rise cannot be said to be mandatory by any means. I consider it likely that final rises (in the narrow sense, cf. Section 4) are primarily attitudinal markers, and not question-forming, as already suggested by a long line of research (Hadding-Koch, 1961; House, 2004, 2005; Myrberg, 2010).

3.2 Phonetics of utterance-final nuclear accent

House (2002) investigated the influence that the intonational features of an utterance-final nuclear accent have on statement vs. question perception in a syntactically declarative sentence (*Hon vill bara flyga*, "She only wants to fly"). The target word *flyga* has A2 and consequently one pitch peak per syllable in Stockholm Swedish. House manipulated two factors: The peak height of *both* accent peaks (130 Hz vs. 160 Hz), and the temporal alignment of the *second*, i.e. post-tonic and post-lexical, accent peak in six steps of 25 ms (the earliest peak was aligned with the CV boundary, while the latest peak was 25 ms before the utterance offset). Participants (mostly speakers of Stockholm Swedish) were asked to judge whether the speaker, who was depicted by an animated computer-generated face[9], intended to make a statement or ask a question. Both pitch accent height and peak alignment influenced question perception: Higher pitch and later peaks lead to increased question perception. There seems to be a category boundary before the two latest pitch peaks, such that even with low accent height, there were more than 60% of question identification for those very late-aligned peaks. Note that a pitch fall that starts 25 or even 50 ms before utterance offset may be too short to be perceptible as a fall, in which case these two conditions would probably have been perceived as rises instead of as rise-falls.

Ambrazaitis, Buanzur, and Niebuhr (2015) showed that there are differences in the *shape* of nuclear accents between Stockholm Swedish statements and questions, such that statements mostly have convex (= fast-rising) accentual rises and questions concave (= slow-rising) accentual rises. Their target words all had A1, so the accent was a post-lexical L*H big accent. They included both canonical (verb-first) polar questions and non-canonical (verb-second) 'repetition questions' and 'disbelieving questions' (roughly echo questions, without and with incredulity). They did not find any differences between the various question types in terms of accent shape, or any differences in the shape of the post-tonic fall to L%. This result is compatible with House (2002, 2003)'s findings on later accent peak location in questions: A given pitch target will be approached earlier with a convex, fast rise and later with a concave, slow rise, all else being equal.

[9]A second sub-experiment investigated the influence of visual cues on sentence mode identification. The hypothesized declarative cues – a smile, narrowing of the eyes and a head nod – did influence perception in the expected direction, while the hypothesized interrogative cues – a slow nod and lowering of the eyebrows – did not.

3.3 Other pitch-related differences

Recall that Gårding (1979) included a flattening of prenuclear accents in her model of Bruce's questions, which resulted from a raising of the lows between accents, cf. Fig. 2. This prenuclear flattening is also visible in all figures that show questions in Bredvad-Jensen (1981), who presents data on the influence of syntax (V2 vs. V1), focus location, and attitude (neutral, polite, determined) on the intonation of statements and questions produced by one speaker from Halmstad (Southern Sweden). It also contains one of the few direct visual comparisons of canonical polar questions and non-canonical declarative questions in the literature (Bredvad-Jensen, 1981, p. 43). Differences between the two are very slight, and probably entirely related to the fact that different lexical material is in focus. Bredvad-Jensen (1984) expands these data with recordings from two more speakers from Southern (Malmö) and Western Sweden (Göteborg). The speaker from Malmö mostly shows a general raising of pitch in the questions, without much flattening. The speaker from Göteborg mostly seems to raise the second peak of the word in focus, possibly in connection with the raising of prenuclear accent lows described by Gårding (1979) (the statements also show comparatively flat prenuclear stretches, albeit much lower than those of the questions).

Horne and Roll (2021), already discussed in Section 3.1.2 with respect to final rises, also investigated early pitch and overall pitch register. To investigate differences in register width between statements and (canonical) questions, they measured maximum F0 on the first word and on the most prosodically prominent word following the first word. For the first word, they report means of 7.6 semitones (relative to 100 Hz) for statements, 10.3 semitones for yes/no questions and 11.1 semitones for *wh*-questions (the difference between both question types and statements being significant). For the most prominent word after the first word, yes/no questions were significantly higher in maximum pitch (12.6 semitones) than both statements (10 semitones) and *wh*-questions (10.9 semitones). Unfortunately, Horne and Roll (2021) do not report these phonetic details for the fragmentary questions, which showed many final rises, so we do not know to which extent final rises and high initial pitch interact.

3.4 Duration & speaking rate

House (2003) (a follow-up study to House, 2002) manipulated three factors in a syntactically declarative sentence (*Hon kan tänka sig åka bil*, "She can see herself going by car"): (i) the position of the pitch peak in the accented vowel of *bil* (A1) in four steps of 50 ms, with the latest step such that the peak is located on the VC boundary and the fall is entirely located within the [l], (ii) the height of the pitch peak (120 Hz vs. 140 Hz), and (iii) whether or not the [a] of *åka* (A2; [ˈoːka]) was lengthened by 150 ms or not (note that *åka* was unaccented and the [a] is in an unstressed syllable, i.e. the lengthening was intended to be perceived as a pre-focal filled pause). Participants (mostly speakers of Stockholm Swedish) were asked to judge whether the speaker was making a statement or asking a question. The results show a general bias towards statement identification, but all factors played a role: The most statement-like conditions had low accent pitch, no pause, and the two earliest accent peaks. The most question-

like conditions (with about 80% question identification) had high accent pitch and the latest accent peak. In this configuration, the presence of a pre-focal pause made little difference (suggesting a ceiling effect), but in the other conditions, the addition of a pause led to a jump towards question identification that was generally at least as large as the difference between low and high accent pitch. In interpreting the results, House suggests that the pre-focal pause indicates hesitation, which in turn is interpreted as uncertainty (which strengthens question perception).

Strömbergsson, House, and Edlund (2012) report substantially faster speaking rate for questions from the same corpus of 600 questions from Strömbergsson, Edlund, and House (2012), when compared to matched statements from the same speakers. However, the authors themselves speculate that this may be an artifact of how questions and non-questions were selected, and that the non-questions happen to contain more non-linguistic material, which artificially reduced speaking rate. If speaking rate was reported in syllables per second, then the mean for non-questions of 3.84 would indeed be quite slow (rather than the questions having genuinely fast speaking rate). Strömbergsson, Edlund, and House (2013) focus on the 48 reported questions in this dataset and find that speech rate is higher in them than in non-reported questions (while perceptually annotated prosodic marking of reportedness was overall less common than anticipated, characterizing only a little more than half of the reported questions).

Seeliger and Repp (2017) also found a slight lengthening of the penultimate syllable in rejecting questions, although this was a stressed syllable, unlike in House (2003). It is thus not clear whether this difference can be related to hesitation like House plausibly argued for his material. The lengthening was independent of focus location, however.

4 Discussion: Late-aligned and late-positioned accents

We have seen that phonological accounts largely agree on the existence of (at least) two distinct boundary tones in Swedish, L% and H%. These are associated with phrase boundaries. They are unrelated to, but must coexist with tone accents in any given phrase. There is also a different perspective on the phenomenon of utterance-final pitch movements, particularly upwards movements, which runs through the literature and is rarely stated outright: If, in questions, pitch range on the prominence tone is widened and pitch peak position is delayed and/or reached after a shallower rise (as suggested by Gårding, 1979; House, 2002, 2003; Ambrazaitis, Buanzur, and Niebuhr, 2015), then an auditory utterance-final rise could well be the result of an accent that has been raised and delayed like this, as opposed to the instantiation of a boundary tone. In other words, the phonetic modification of (default) sentence prosody that seems to characterize Swedish questions might perceptually 'turn' a tone accent rise into something resembling a question rise (in the terminology laid out in Section 2.3), even in varieties that generally have L% boundary tones.

This idea can be seen in the background of all of House's work on question prosody (House, 2002, 2003, 2004, 2005). I want to highlight just one

specific hypothesis (House, 2004, p. 59): House suggests that speakers might choose an indirect question like *Vet du var Kulturhuset ligger?* ("Do you know where Kulturhuset is?"), in which the nuclear accent lies on *Kulturhuset* in non-final position, over a direct question like *Var ligger Kulturhuset?* ("Where is Kulturhuset?"), which has the nuclear accent in final position, specifically to avoid any possibility of producing a final rise, and thus being heard as socializing instead of as information-seeking. Fig. 4 shows a schematic contour for the Stockholm Swedish pronunciation of *Kulturhuset* in utterance-final position, with the L*H prominence tone close to the end of the utterance. If, in Stockholm Swedish questions, the H of the prominence tone is realized as phonetically extra high and/or extra late, that moves it even closer to the end of the utterance, which might make it more likely to sound utterance-final in questions than in statements. If the word is instead followed by *ligger*, with a HL* small A1 accent, there is an additional *accentual* fall before the L% boundary tone, which presumably makes it less likely for the prominence tone (which is now also further from the end of the utterance) to be (mis)perceived as a question rise.

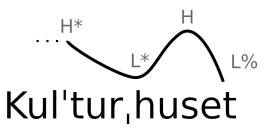

Figure 4: Schematic contour for the compound word *Kulturhuset* ("the culture house") in Stockholm Swedish, utterance-final position.

To my knowledge, the first observation of late accent placement being conducive to question interpretation is found in Johansson (1978). In a perception study with a resynthesized nonsense stimulus with four syllables and two pitch peaks (with listeners from Northern Sweden, where the dialect is similar to Central Swedish in terms of word prosody, but distinct in terms of sentence prosody, cf. Bruce, 2010), she manipulated (among other factors) the temporal location of the highest pitch peak inside the stimulus. She summarizes the results as follows: "Contours were judged as statement intonation if they had their most prominent frequency peak in their first half, and contours were judged as question intonation if their peak fell in the second half of the contour" (Johansson, 1978, p. 52, my translation). Crucially, of the four most question-like contours shown in her Fig. 3.7 on p. 52, the contour with the latest pitch peak has a visibly cut-off fall, which makes it possible that it was perceived as a rise.

Huhtamäki (2015a, p. 61) goes so far as to suggest that "a late and prominent accent of this type has been perceived as a rise in older auditory analyses of question prosody in Swedish" (my translation). It must be noted, however, that she is referring to utterance-final nuclear accents in general, not phonetically extra late and/or high accent peaks, and Swedish nuclear accents are right-aligned in broad focus statements and questions alike – so everything else being equal, these (mis)perceived rises should be equally likely in statements and questions on Huhtamäki's reasoning.

5 Summary

Overall, we find clear evidence for phonetic modification of tone accents in Swedish questions, such that pitch peaks are raised and/or delayed relative to statement intonation (particularly pitch peaks of *post*-lexical prominences). We find little to no evidence of phonological, categorical distinctions between question and statement intonation in Swedish. There is evidence for an interesting interaction of word and sentence prosody: phonetically extra high and/or late accent peaks might be prone to be (mis)heard as utterance-final, i.e. sentence-level, rises.

Regarding the issue of whether non-canonical questions have 'more' or 'less' question prosody than canonical ones, we can tentatively conclude that this distinction is also not categorical. Gårding (1979)'s model assumed a continuum of question prosody, of which non-canonical questions have more than canonical ones (but this continuum was a phonetic modification of statement prosody). Bredvad-Jensen (1984)'s dataset showed little difference between canonical polar questions and non-canonical declarative questions. Ambrazaitis, Buanzur, and Niebuhr (2015) only found (phonetic) distinctions between statements and questions, but not between canonical and non-canonical questions. Overall, however, much more work on the prosody of Swedish questions is needed, particularly also on non-pitch parameters such as voice quality, before a typology of Swedish question prosody is even close to complete.

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